MSDS 452: Graphical, Network, and Causal Models

Checkpoint-C

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Causality analysis of student performance factors

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Abstract

This study investigates the causal mechanisms of student academic performance by applying Causal Bayesian Networks to a dataset of secondary math students from two Portuguese schools. Moving beyond simple predictive correlation, the research employed a hybrid strategy combining automated Causal Structural Discovery (CSD) algorithms, such as PC Stable and GES, with manual refinement based on domain expertise and iterative d-separation testing. The resulting 11-variable Directed Acyclic Graph (DAG) identifies critical causal pathways, such as the influence of student motivation on study time and the direct impact of parental education on academic failures. The resulting causal structure is a robust framework for future intervention and counterfactual analysis.

Keywords: Biomarker Discovery, Causal Bayesian Networks, Causal Structural Discovery, Intervention, Counterfactual

1.0 Introduction

Understanding the key drivers of student academic performance is the main theme of educational research and policy, with profound implications in shaping effective public spending, providing equal opportunities for the next generation, and ultimately fostering a competitive workforce that powers the economy. A large body of literature has established strong correlations between student success and factors like socio-economic status, past performance, and school environment. While these factors are good predictors of student performance, the underlying statistical models cannot establish causality, let alone draw conclusions on the causal effect of intervention policies. Costly randomized controlled experiments can be performed as a remedy, but they are inefficient and often difficult to set up. This study aims to move beyond predictive modeling to investigate the causal mechanisms that determine student achievement using causal inference.

I used a dataset of student math performance measured in grades in secondary education of two Portuguese schools, available from the UCI Machine Learning Repository (Cortez 2008). It contains 649 instances of 30 attributes, capturing a comprehensive profile of students, including their demographic background (e.g., sex, age, parent's education and jobs), social characteristics (e.g., family size, parental cohabitation status), and school-related features (e.g., study time, past failures, extra support). Using causal inference methods, I can identify potential confounders (like family background, measures of motivation) and investigate the causal effect treatment variables, such as extra educational support, intervention on school absence, or promoting student motivation for higher education).

2.0 Literature review

The heart of causal inference is modeling variable relationships with Directed Acyclic Graphs (DAG). The nodes represent variables and the directed edges represent the functions mapping one variable to another. The graph model imposes constraints on the variables so that each variable is independent of its non-descendants given its parents (Markov conditions). The parent (P) to child (C) relationship between nodes expresses the dependency of their probability distribution, $P(P,C) = P(C \mid P)P(P)$. By applying the chain rule of probability, the joint probability of all variables, $P(data|model)$, can be calculated as the product of the independent Markov kernels. Like all other machine learning algorithms, causal inference aims to find the optimal causal model that best fits the data, that is, maximizes the likelihood or a posteriori. To achieve this, we need to establish the DAG and learn the parameters that determine the conditional dependencies between all P and C.

The initial causal model can be constructed manually based on human intuition and domain knowledge, through causal structure discovery (CSD) algorithms, or a combination of both. The causal relationships between variables induce independence or conditional independence for d-separations, defined as the path between nodes blocked by colliders (in A→B←C, A and C are separated) or by conditioning on the middle nodes of a chain (A→B→C) or a fork (A←B→C) (Pearl, Glymour and Jewell 2016). By performing independence tests, the current model is either accepted or rejected (Ness 2025). Modifications can be made directly to edges of the failed tests, and further independence tests are carried out for each new model until the irrefutable model is reached. Parameter fitting

When human knowledge in an area is limited or biased, and as the dimensionality of datasets grows, CSD provides a viable approach to generate an initial DAG for subsequent testing. The goal of CSD is to find the most optimal Bayesian network with a given set of nodes that represent dependent features. The challenge of this optimization problem is not only that the number of possible network structures grows super-exponentially with the number of nodes, but also that the solution space is discrete Causal discovery methods are broadly categorized into constraint-based and score-based algorithms, typically applied when the sample size significantly exceeds the number of observed variables (usually$N \gg p$, where$p < 30$). Constraint-based approaches, such as the PC (Peter-Clark) algorithm (Spirtes, Glymour and Scheines 2001) and its latent-variable variants (Spirtes, Meek and Richardson 1995), infer edge presence and direction by strictly utilizing statistical dependence and conditional independence tests. In contrast, score-based algorithms identify the optimal structure by maximizing a Bayesian scoring criterion that measures the local goodness-of-fit for each node relative to its parents. Notable among these is Greedy Equivalence Search (GES) (Chickering 2002), a two-phase process that drastically reduces the search space by navigating Markov equivalence classes rather than individual DAGs. Similar exact search method introduced by Silander and Myllymäki (2006) follows the steps of identifying most likely parents of each node and the best sink node, and reordering the sequences of causal chain to maximize a Bayesian scoring criterion..

CSD algorithms rely on heavy theoretical assumptions, such as causal faithfulness or Meek conjecture, and show limitations in finalizing edge directions, handling latent variable, and sensitivity to data quality and size. Therefore, the strategy for causal modeling should be utilizing causal discovery to narrow down candidate graphs to be used in subsequent causal inference testing, while blending in human oversight and domain expertise.

The true power of causal models lies in their ability to infer causal effects through intervention and counterfactual analyses. (more on this in the final report).

3.0 Methods

3.1 Exploratory Data Analysis and data preparation

I first exclude the age variable, since all cases are in secondary education. The longitudinal grades within one school year (G1, G2, G3) were summed to generate a total `grade` variable.

Two sets of numerical variables that showed strong correlation (Figure 1) were aggregated by summing: `Fedu` (father's education) and `Medu` (mother's education) to generate `Pedu`; `Dalc` (workday alcohol consumption) and `Walc` (weekend alcohol consumption) to generate `alc`. This resulted in 27 predictor variables.

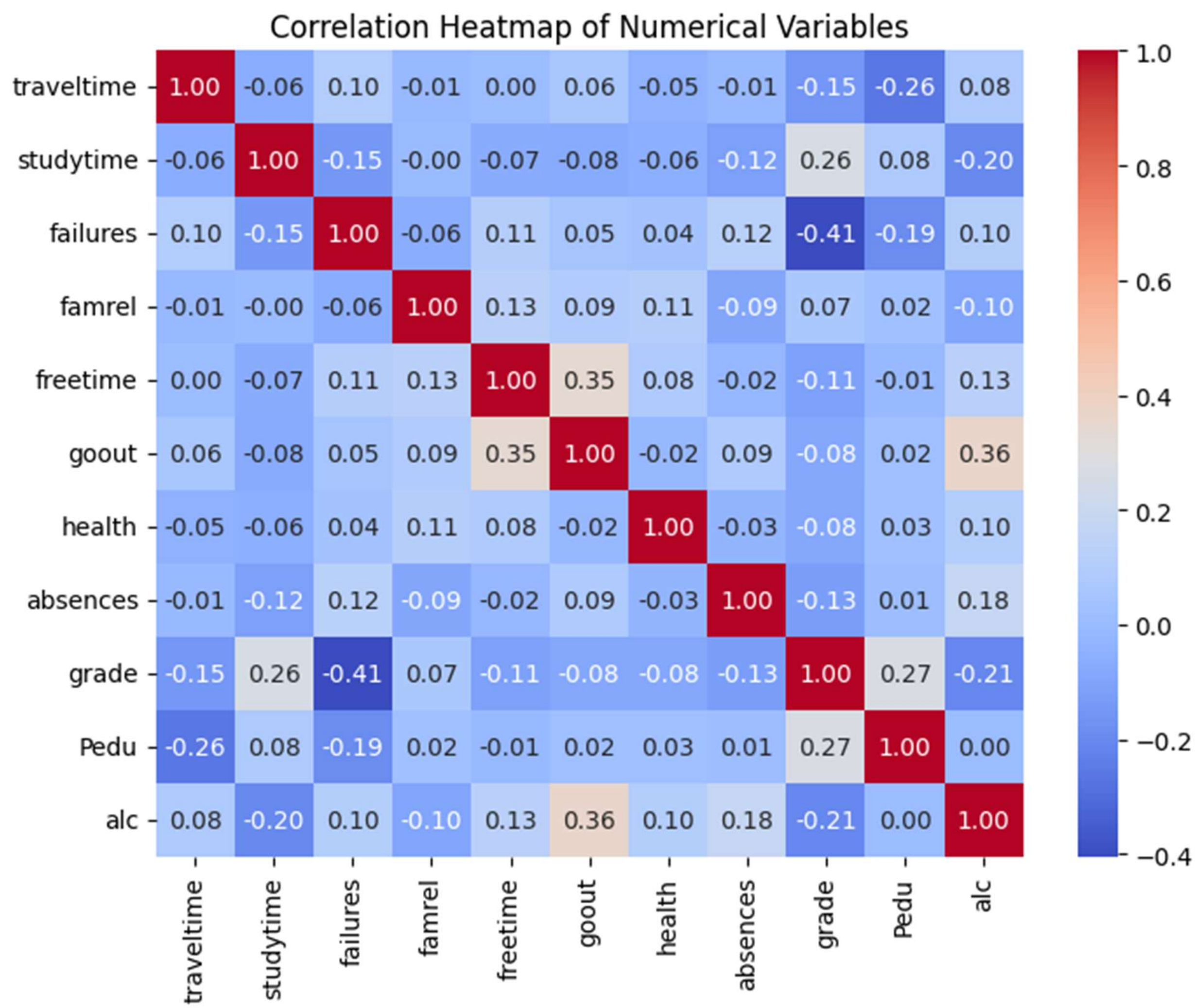

Figure 1 Correlation matrix of numerical features

A random forest regressor was then trained to identify important features. I prioritized features that exhibit a consistent and monotonic relationship with the model's output, as evidenced by clear color separation along the SHAP (Shapley Additive exPlanations) value axis. Specifically, I selected variables where high feature values (red) are distinctly clustered on one side of the center line and low values (blue) on the other (Figure 2), indicating a stable contribution to the prediction, with some tolerance of noise for features that are in the top 10 of important features. The selected features are:

```
top_features = ['failures', 'alc', 'Pedu', 'absences', 'higher',
        'studytime', 'school', 'schoolsup', 'address', 'sex']
```

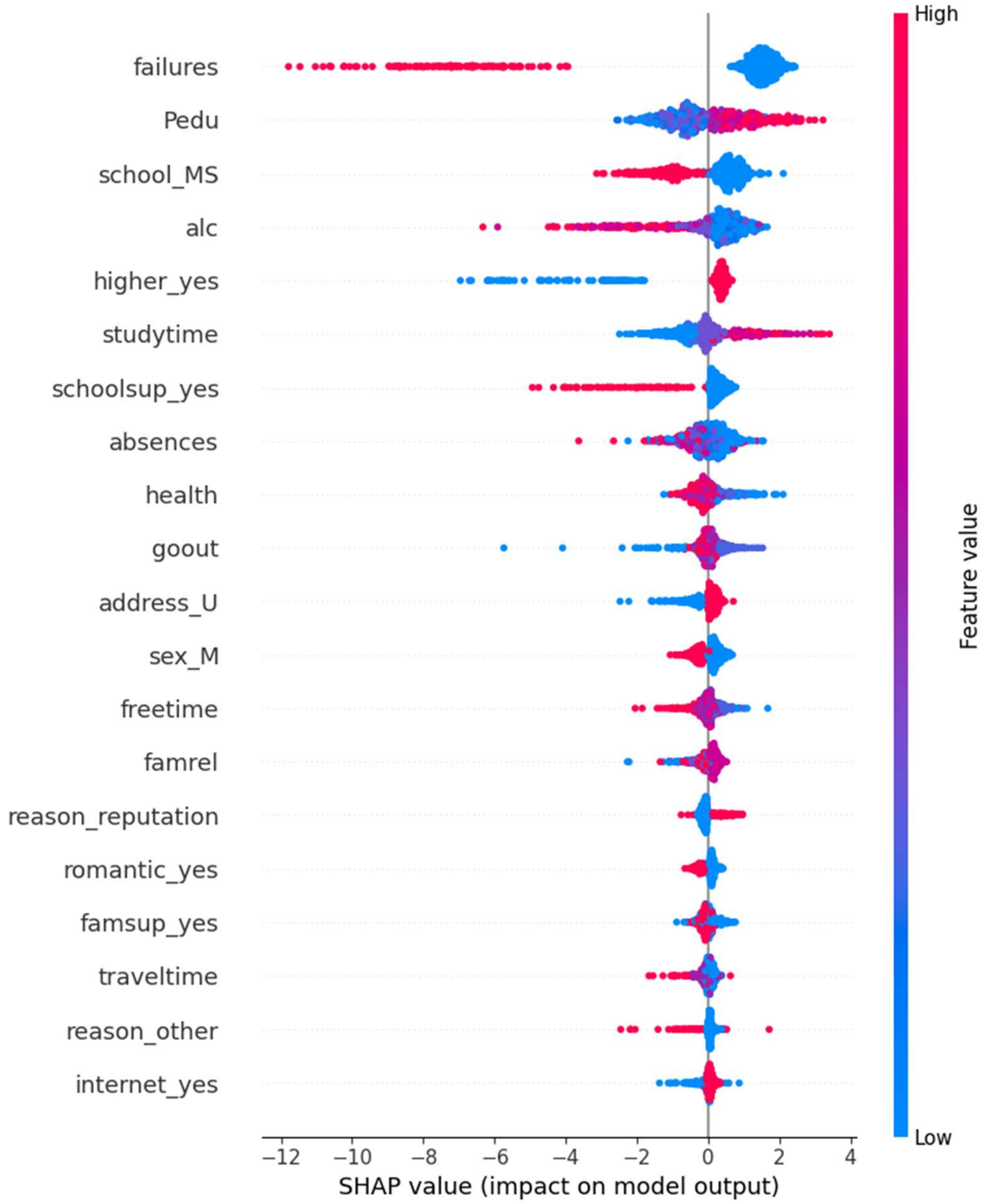

Figure 2 SHAP summary plot of important features

The continuous grade variable was discretized to generate grade_group with three categories based on its distribution: Low (bottom 20th percentile), High (top 20th percentile), and Mid (the remaining 60%).

Variable absences was also discretized to generate absences_group with two categories: normal (0-5), high (>5).

Standardizing the data types across all variables ensures compatibility with specific conditional independence tests for constraint-based algorithms and scoring functions for score-based methods.

3.2 Causal discovery algorithms

I used the Peter-Clark (PC) Stable algorithm and Greedy Equivalent Search (GES) algorithm implemented in causal-learn python module (Zheng, et al. 2024). The independent test for PC was Chi-squared test. The scoring criterion in GES for discrete data is Bayesian Dirichlet equivalent uniform (BDeu) for Dirichlet distribution with uniform prior on graph structure.

3.3 d-separations and independence test

The d-separations were calculated using get_independencies() function from the pgmpy module v1.0.0 (Ankan and Textor 2024).

The appropriate independence test for categorical variables is chi-squared test. A p-value threshold of 0.05 was chosen to determine dependences between variables: $p < 0.05$, dependence exists and the model is refuted. No family-wise error correction was used.

3.4 Intervention assessment

3.5 Counterfactual analysis

4.0 Results

4.1 Constructing the DAG

The initial PC algorithm output (Figure 3) failed to incorporate absences_group, despite significant dependencies identified via pairwise chi-squared tests (Figure 4). Furthermore, the algorithm produced undirected edges between higher, studytime, and grade_group. Given that higher (student motivation) is a foundational driver of academic behavior, it should have been positioned closer to the root of the structure. To address these inferential errors, I introduced the following structural priors to the learning process:

- `higher` → `studytime` & `higher` → `grade_group`: Directionality reflects that motivation dictates academic effort and outcomes.
- `address` → `school`: A constraint based on geographical and administrative logic.
- Edge Exclusion (`grade` ↮ `failures`): I enforced the absence of a direct link, assuming their correlation is mediated by latent or underlying factors rather than a direct causal path.

The outcome graph (Figure 3, bottom) corrected the above errors but was still inconsistent with the pair-wise independence tests (Figure 4).

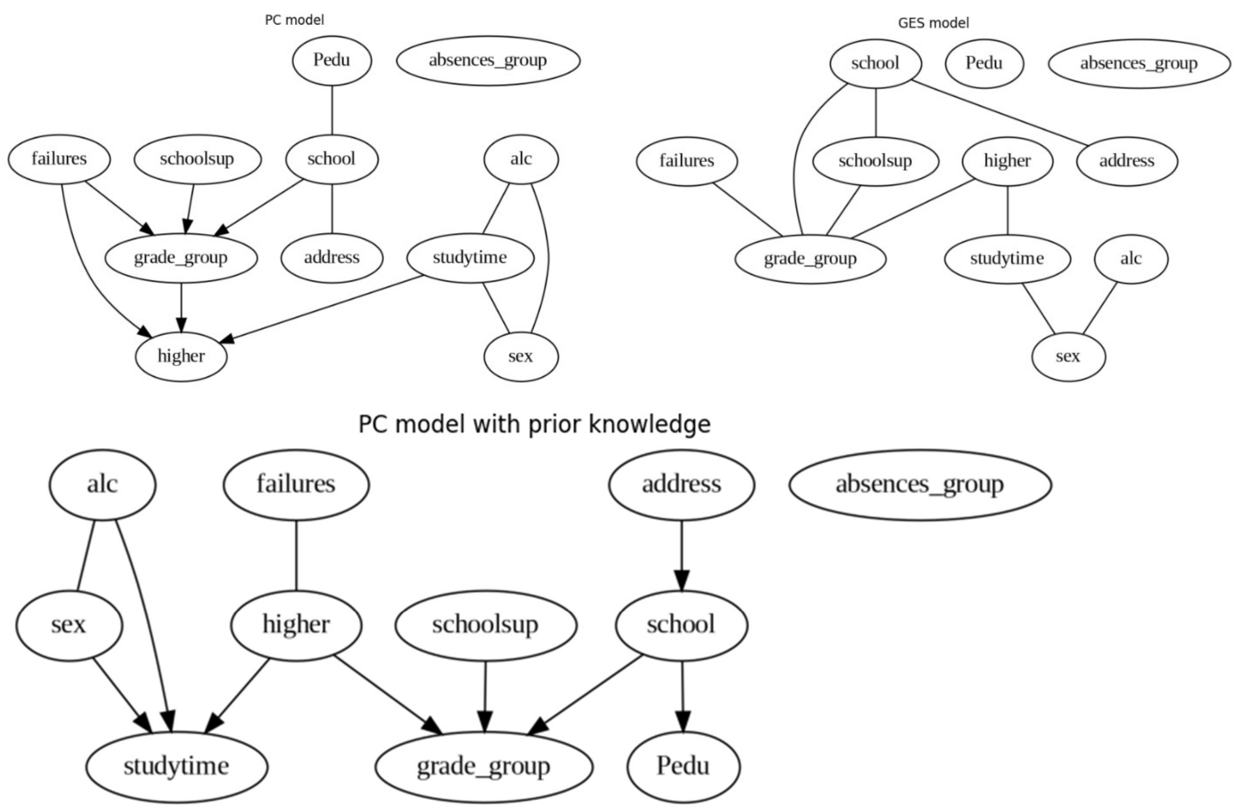

Figure 3 Top left, output DAG of PC. Top right, output DAG of GES. Bottom, output of constrained PC

The GES algorithm produced a Completed Partially Directed Acyclic Graph (CPDAG), as it searches across Markov equivalence classes rather than individual DAGs. While the algorithm was unable to orient all edge directions from the data alone (Figure 3, top right), it confirmed the structural dependencies within two key clusters: (`alc`, `studytime`, `sex`) and (`higher`, `studytime`, `grade_group`). These identified triads served as a helpful starting point for manual structural refinement.

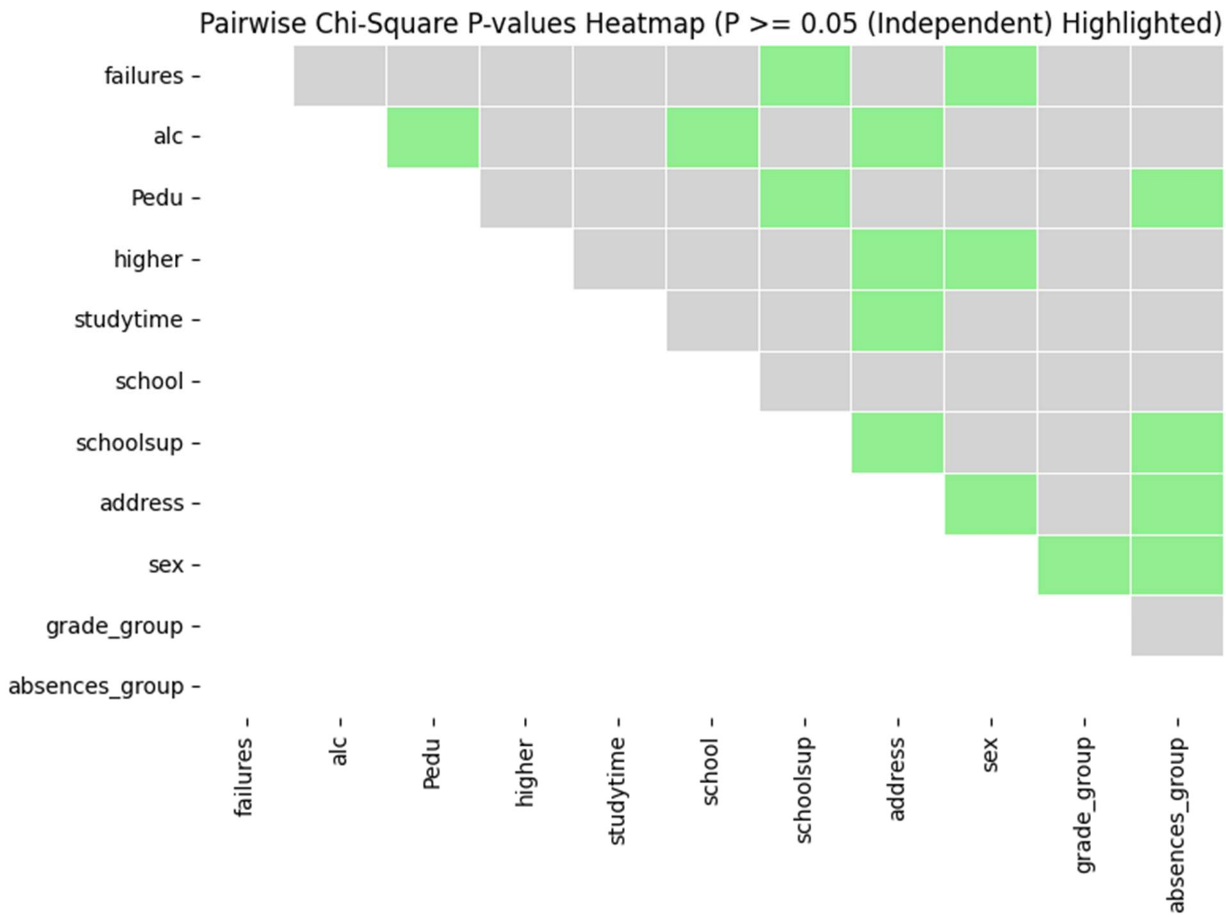

Figure 4 Pair-wise chi-squared test. Independences (p>=0.05) were highlighted in green.

Starting with the skeleton provided by the constrained PC algorithm (Figure 3) and initial independence tests (Figure 4), I manually constructed a candidate Directed Acyclic Graph (DAG) (Figure 5, left). Using the `pgmpy` library, I iteratively performed independence tests of d-separations to identify discrepancies between the model and the data. After refining the structure to resolve these falsified dependencies, I arrived at a final model (Figure 5, right) that is fully consistent with the observed independence constraints. It reveals some interesting causal relationships such as, `Pedu` → `sex`, `sex` → `school`, and `Pedu` has a direct effect on `failure` that is not through other factors, `school` also have a direct effect on grades besides the path `school` → `studytime` → `grade`.

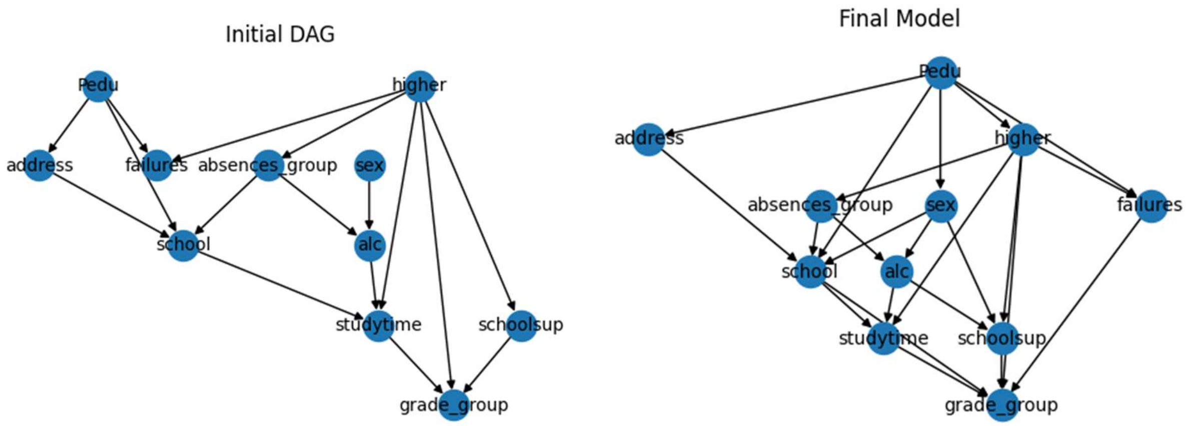

Figure 5 Left, initial manually constructed DAG. Right, final model that is consistent with the data.

4.2 Model fitting

4.3 Intervention analysis

4.4 Counterfactual analysis

5.0 Conclusion

The causal discovery algorithm did not successfully learn the perfect graph, although it provided a good initial guess of some dependencies in the data. With this 11-variable problem, human-guided modifications can achieve consistency with the data. Interestingly, the final graph appears denser than the outputs of causal discovery algorithms, suggesting underlying sparsity restrictions in the tested CSD algorithms.

The fact that the null hypotheses of independence tests were not rejected for the final graph structure does not imply the model is correct. It only suggests that there is no evidence in the data to refute it. Further evaluations after learning the parameters are necessary to further refinement.

Next, I will perform model fitting, do operations and counterfactual analysis to infer effective intervention policies for improving math grade in secondary education.

References

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